1. In the following, upper triangular matrix is :

निम्न में से ऊपरी त्रिभुजाकार मैट्रिक्स है :

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & 3 \end{bmatrix}$ (B) $\begin{bmatrix} 5 & 4 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}$ (D) $\begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 0 & 0 \end{bmatrix}$

Ans. B

Sol. An upper triangular matrix is one where all entities below the main diagonal are zero.

2. In the following, singular matrix is :

निम्न में से अव्युत्क्रमणीय मैट्रिक्स है :

(A) $\begin{bmatrix} 2 & 3 \\ 1 & 3 \end{bmatrix}$ (B) $\begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$

Ans. D

Sol. $|A| = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 2 \times 6 - 3 \times 4 = 0$

3. If $A = \begin{bmatrix} 5 & 2 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$, then $|2A - 3B|$ equals :

यदि $A = \begin{bmatrix} 5 & 2 \\ 1 & 0 \end{bmatrix}$ तथा $B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$ तो $|2A - 3B|$ बराबर है :

(A) 77 (B) – 53 (C) 53 (D) – 77

Ans. B

Sol. $2A - 3B = 2\begin{bmatrix} 5 & 2 \\ 1 & 0 \end{bmatrix} - 3\begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$

$$= \begin{bmatrix} 4 & -5 \\ -13 & 3 \end{bmatrix}$$

$= 12 - 65 = -53$

4. If a matrix B is obtained by multiplying each element of a matrix A of order 2×2 by 3, then relation between A and B is :

यदि मैट्रिक्स A जिसकी कोटि 2 x 2 के प्रत्येक अवयव को 3 से गुणा करने पर मैट्रिक्स B प्राप्त होता हो तो A तथा B में निम्न संबंध होगा :

(A) $A = 3B$ (B) $3A = B$ (C) $9A = B$ (D) $A = 9B$

Ans. B

Sol. $3A = B$

5. If $\begin{bmatrix} 5 & 7 \\ x & 1 \\ 2 & 6 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ -3 & 5 \\ 2 & y \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 4 & -4 \\ 0 & 4 \end{bmatrix}$ then :

यदि $\begin{bmatrix} 5 & 7 \\ x & 1 \\ 2 & 6 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ -3 & 5 \\ 2 & y \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 4 & -4 \\ 0 & 4 \end{bmatrix}$ तो :

(A) $x = 1, y = -2$ (B) $x = -1, y = 2$ (C) $x = 1, y = 2$ (D) $x = -1, y = -2$

Ans. C

Sol. $x = 1, y = 2$

6. If $A + B = \begin{bmatrix} 7 & 4 \\ 8 & 9 \end{bmatrix}$ and $A - B = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ then the value of A is :

यदि $A + B = \begin{bmatrix} 7 & 4 \\ 8 & 9 \end{bmatrix}$ और $A - B = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ तो A का मान होगा :

(A) $\begin{bmatrix} 3 & 1 \\ 4 & 3 \end{bmatrix}$ (B) $\begin{bmatrix} 4 & 3 \\ 4 & 6 \end{bmatrix}$ (C) $\begin{bmatrix} 6 & 2 \\ 8 & 6 \end{bmatrix}$ (D) $\begin{bmatrix} 7 & 6 \\ 8 & 12 \end{bmatrix}$

Ans. B

Sol. If $A+B=\begin{bmatrix}7 & 4\\ 8 & 9\end{bmatrix}$ and $A-B=\begin{bmatrix}1 & 2\\ 0 & 3\end{bmatrix}$ then the value of A is $\begin{bmatrix}4 & 3\\ 4 & 6\end{bmatrix}$

7. The scalar matrix is :

(A) $\begin{bmatrix}-1 & 3\\ 2 & 4\end{bmatrix}$ (B) $\begin{bmatrix}0 & 3\\ 2 & 0\end{bmatrix}$ (C) $\begin{bmatrix}4 & 0\\ 0 & 4\end{bmatrix}$ (D) None of these

अदिश मैट्रिक्स है :

(A) $\begin{bmatrix}-1 & 3\\ 2 & 4\end{bmatrix}$ (B) $\begin{bmatrix}0 & 3\\ 2 & 0\end{bmatrix}$ (C) $\begin{bmatrix}4 & 0\\ 0 & 4\end{bmatrix}$ (D) इनमें से कोई नहीं

Ans. C

Sol. The scalar matrix is a diagonal matrix, in which the diagonal contains the same element.

The scalar matrix is obtained by the product of the identity matrix and a scalar number.

8. For any square matrix $A=[a_{ij}]$, $a_{ij}=0$, when $i\neq j$, then A is :

(A) unit matrix (B) scalar matrix (C) diagonal matrix (D) None of these

यदि किसी वर्ग मैट्रिक्स $A=[a_{ij}]$, $a_{ij}=0$, के लिए जबकि $i\neq j$, तो A होगा :

(A) इकाई मैट्रिक्स (B) अदिश मैट्रिक्स (C) विकर्ण मैट्रिक्स (D) इनमें से कोई नहीं

Ans. C

Sol. We have, $a_{ij}=0$ when $i\neq j$ That mean all the non diagonal elements of the matrix are zero, So the matrix is,

$$A\begin{bmatrix}a_{11} & 0 & 0 & \ldots\ldots\ldots\ldots & 0\\ 0 & a_{22} & 0 & \ldots\ldots\ldots & 0\\ 0 & 0 & a_{33} & \ldots\ldots & 0\\ & & & & \\ & & & & \\ 0 & 0 & 0 & \ldots\ldots\ldots\ldots & 0\end{bmatrix},$$

where n is order of the matrix clearly A is a diagonal matrix

9. In the following, diagonal matrix is :

निम्न में से विकर्ण मैट्रिक्स है :

(A) $\begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$

Ans. D

Sol. A diagonal matrix is a matrix (usually a square matrix in which the entries outside the main diagonal are all zero.)

10. If $\begin{bmatrix} x & 3x-y \\ zx+z & 3y-w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$, then :

(A) $x = 3, y = 7, z = 1, w = 14$ (B) $x = 3, y = -5, x = -1, w = -4$
(C) $x = 3, y = 6, z = 2, w = 7$ (D) None of these

यदि $\begin{bmatrix} x & 3x-y \\ zx+z & 3y-w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$, तो :

(A) $x = 3, y = 7, z = 1, w = 14$ (B) $x = 3, y = -5, x = -1, w = -4$
(C) $x = 3, y = 6, z = 2, w = 7$ (D) इनमें से कोई नहीं

Ans. A

Sol. Given that $\begin{bmatrix} x & 3x-y \\ zx+z & 3y-w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$

comparing both sides :

$x = 3$

$3x - y = 2$

$\Rightarrow y = 9 - 2 = 7$

$zx + z = 4$

$\Rightarrow z = \frac{4}{3+1} = 1$

$3y - w = 7$

$\Rightarrow w = 21 - 7 = 14$

11. If $A = \begin{bmatrix} 1 & -5 & 7 \\ 0 & 7 & 9 \\ 11 & 8 & 9 \end{bmatrix}$, then the value of tr (A) is :

यदि $A=\begin{bmatrix}1 & -5 & 7\\ 0 & 7 & 9\\ 11 & 8 & 9\end{bmatrix}$ तो tr (A) का मान होगा :

(A) 17 (B) 25 (C) 3 (D) 12

Ans. A

Sol. $\text{tr}(A)=1+7+9=17$

12. If $A=\begin{bmatrix}\sin^2\alpha & \sec^2\alpha\\ \text{cosec}^2\alpha & 1/2\end{bmatrix}$ and $B=\begin{bmatrix}\cos^2\alpha & -\tan^2\alpha\\ -\cot^2\alpha & 1/2\end{bmatrix}$, then A + B is equal to :

यदि $A=\begin{bmatrix}\sin^2\alpha & \sec^2\alpha\\ \text{cosec}^2\alpha & 1/2\end{bmatrix}$ and $B=\begin{bmatrix}\cos^2\alpha & -\tan^2\alpha\\ -\cot^2\alpha & 1/2\end{bmatrix}$ तो A + B बराबर है :

(A) $\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$ (B) $\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$ (C) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ (D) $\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$

Ans. B

Sol. Given, $A=\begin{bmatrix}\sin^2\alpha & \sec^2\alpha\\ \text{cosec}^2\alpha & \frac{1}{2}\end{bmatrix}$ and $B=\begin{bmatrix}\cos^2\alpha & -\tan^2\alpha\\ -\cot^2\alpha & \frac{1}{2}\end{bmatrix}$

$\therefore A+B=\begin{bmatrix}\sin^2\alpha & \sec^2\alpha\\ \sec^2\alpha & \frac{1}{2}\end{bmatrix}+\begin{bmatrix}\cos^2\alpha & -\tan^2\alpha\\ -\cot^2\alpha & \frac{1}{2}\end{bmatrix}=\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$

13. If $A=\begin{bmatrix}3 & 5\\ -2 & 4\end{bmatrix}$ and $B=\begin{bmatrix}1 & -3\\ 2 & -7\end{bmatrix}$, then 2A – 3B is equal to :

(A) $\begin{bmatrix}3 & -19\\ 10 & 29\end{bmatrix}$ (B) $\begin{bmatrix}3 & 19\\ -10 & 29\end{bmatrix}$ (C) $\begin{bmatrix}-3 & 19\\ 10 & 29\end{bmatrix}$ (D) None of these

यदि $A=\begin{bmatrix}3 & 5\\ -2 & 4\end{bmatrix}$ तथा $B=\begin{bmatrix}1 & -3\\ 2 & -7\end{bmatrix}$ तो 2A – 3B बराबर है :

(A) $\begin{bmatrix} 3 & -19 \\ 10 & 29 \end{bmatrix}$ (B) $\begin{bmatrix} 3 & 19 \\ -10 & 29 \end{bmatrix}$ (C) $\begin{bmatrix} -3 & 19 \\ 10 & 29 \end{bmatrix}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $2A - 3B = 2\begin{bmatrix} 3 & 5 \\ -2 & 4 \end{bmatrix} - 3\begin{bmatrix} 1 & -3 \\ 2 & -7 \end{bmatrix} = \begin{bmatrix} 3 & 19 \\ -10 & 29 \end{bmatrix}$

14. If $\begin{bmatrix} 16 & -10 \\ -6 & 8 \end{bmatrix} + 2B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, then B is equal to :

यदि $\begin{bmatrix} 16 & -10 \\ -6 & 8 \end{bmatrix} + 2B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, तो B बराबर है :

(A) $\begin{bmatrix} 8 & -5 \\ -3 & 4 \end{bmatrix}$ (B) $\begin{bmatrix} -8 & 5 \\ -3 & 4 \end{bmatrix}$ (C) $\begin{bmatrix} -8 & 5 \\ 3 & -4 \end{bmatrix}$ (D) $\begin{bmatrix} 8 & -5 \\ 3 & -4 \end{bmatrix}$

Ans. C

Sol. $\begin{bmatrix} 16 & -10 \\ -6 & 8 \end{bmatrix} + 2B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$2B = \begin{bmatrix} -16 & 10 \\ 6 & -8 \end{bmatrix}$

$B = \begin{bmatrix} -8 & 5 \\ 3 & -4 \end{bmatrix}$

15. If $X = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ and $3X - \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$, then the value of a is :

यदि $X = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ तथा $3X - \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$ तो a का मान है :

(A) – 2 (B) 0 (C) 2 (D) 1

Ans. C

Sol. $3x = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}$

$x = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

$a = 2$

16. If A and B are matrices of order $m \times n$ and $n \times n$ respectively, then which of the following are defined :

यदि A तथा B क्रमशः $m \times n$ तथा $n \times n$ कोटि के मैट्रिक्स हों, तो निम्न में से कौनसा परिभाषित है :

(A) AB, BA (B) AB, A^2 (C) A^2, B^2 (D) AB, B^2

Ans. D

Sol. In order for matrix multiplication to be defined, the number of columns in the first matrix must be equal to the number of rows in the second matrix.

1. If $A=\begin{bmatrix}3 & 1\\ 7 & 5\end{bmatrix}$ and $A^2+kI=8A$, then k equals :

   यदि $A=\begin{bmatrix}3 & 1\\ 7 & 5\end{bmatrix}$ तथा $A^2+kI=8A$, तो k बराबर है :

   (A) 4 (B) 8 (C) 1/4 (D) 1/16

Ans. B

Sol. $A^2=\begin{bmatrix}3 & 1\\ 7 & 5\end{bmatrix}\begin{bmatrix}3 & 1\\ 7 & 5\end{bmatrix}=\begin{bmatrix}16 & 8\\ 56 & 32\end{bmatrix}$

$kI=\begin{bmatrix}24 & 8\\ 56 & 40\end{bmatrix}-\begin{bmatrix}16 & 8\\ 56 & 32\end{bmatrix}$

$k\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=\begin{bmatrix}8 & 0\\ 0 & 8\end{bmatrix}$

$k=8$

2. If A,B,C are matrices of order 1×3, 3×3 and 3×1 respectively, the order of ABC will be :

   यदि A,B,C क्रमशः 1×3, 3×3 तथा 3×1 कोटि के मैट्रिक्स हों, तो ABC होगा :

   (A) 3×3 (B) 1×3 (C) 1×1 (D) 3×1

Ans. C

Sol. Order of AB = 1×3

Order of ABC = 1×1

3. If $A=\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}$, $B=\begin{bmatrix}-1 & 1\\ 1 & -1\end{bmatrix}$, then :

   यदि $A=\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}$, $B=\begin{bmatrix}-1 & 1\\ 1 & -1\end{bmatrix}$, तो :

(A) $AB = 0$ (B) $AB = 2I$ (C) $BA = 0$ (D) $B^2 = I$

Ans. A

Sol. $AB = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

4. The root of the equation $[x\ 12]\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}\begin{bmatrix} x \\ -1 \\ 1 \end{bmatrix} = 0$ is :

समीकरण $[x\ 1\ 2]\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}\begin{bmatrix} x \\ -1 \\ 1 \end{bmatrix} = 0$ का मूल है :

(A) 1/3 (B) – 1/3 (C) 0 (D) 1

Ans. A

Sol. To find the root of the equation.

Given, $\begin{bmatrix} x12 \end{bmatrix}\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}\begin{bmatrix} x \\ -1 \\ 1 \end{bmatrix} = 0$.

$\begin{bmatrix} 3x + 2x + 1 \end{bmatrix} \times \begin{bmatrix} x \\ -1 \\ 1 \end{bmatrix} = 0$

$3x - 1 = 0$

$x = \frac{1}{3}$

5. If A, B are two matrices such that $A+B=\begin{bmatrix}1 & -2\\ 2 & 4\end{bmatrix}, A-B=\begin{bmatrix}3 & 2\\ -2 & 0\end{bmatrix}$ then AB equals :

यदि A, B दो मैट्रिक्स इस प्रकार से हों कि $A+B=\begin{bmatrix}1 & -2\\ 2 & 4\end{bmatrix}, A-B=\begin{bmatrix}3 & 2\\ -2 & 0\end{bmatrix}$ तो AB बराबर है : :

(A) I (B) $\begin{bmatrix}-2 & -4\\ 4 & 4\end{bmatrix}$ (C) $\begin{bmatrix}-2 & -4\\ 3 & 2\end{bmatrix}$ (D) $\begin{bmatrix}2 & 4\\ -4 & 4\end{bmatrix}$

Ans. B

Sol. $2A=\begin{bmatrix}4 & 0\\ 0 & 4\end{bmatrix}$

$A=\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}$

$B=\begin{bmatrix}-1 & -2\\ 2 & 2\end{bmatrix}$

$AB=\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}\begin{bmatrix}-1 & -2\\ 2 & 2\end{bmatrix}=\begin{bmatrix}-2 & -4\\ 4 & 4\end{bmatrix}$

6. If $A=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$, then A^2 equals :

यदि $A=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$, तो A^2 बराबर है :

(A) $\begin{bmatrix}\cos 2\alpha & \sin 2\alpha\\ -\sin 2\alpha & \cos 2\alpha\end{bmatrix}$ (B) $\begin{bmatrix}\cos 2\alpha & -\sin 2\alpha\\ \sin 2\alpha & \cos 2\alpha\end{bmatrix}$

(C) $\begin{bmatrix}\sin 2\alpha & \cos 2\alpha\\ -\cos 2\alpha & \sin 2\alpha\end{bmatrix}$ (D) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

Ans. A

Sol. Since

$$A^2 = A.A = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}.$$

7. For matrices A and B, AB = 0, then :

(A) A = 0 or B = 0 (B) A = 0 and B = 0

(C) It is not necessary that A = 0 or B = 0 (D) All above statements are wrong

मैट्रिक्स A तथा B के लिऐ, AB = 0, तो :

(A) A = 0 या B = 0 (B) A = 0 तथा B = 0

(C) यह आवश्यक नहीं है कि A = 0 या B = 0 (D) उपरोक्त सभी कथन असत्य हैं

Ans. C

Sol. It is not necessary that, if AB = 0 then A = 0 or B = 0

Take example, $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

Clearly $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ but $A, B \neq 0$

8. If $A = \begin{bmatrix} -1 & 0 & 2 \\ 3 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 5 \\ 2 & 7 \\ 3 & 10 \end{bmatrix}$, then :

(A) AB and BA both exist (B) AB exists but not BA

(C) BA exists but not AB (D) both AB and BA do not exist

यदि $A = \begin{bmatrix} -1 & 0 & 2 \\ 3 & 1 & 2 \end{bmatrix}$ तथा $B = \begin{bmatrix} -1 & 5 \\ 2 & 7 \\ 3 & 10 \end{bmatrix}$, तो :

(A) AB तथा BA दोनों विद्यमान हैं (B) AB विद्यमान है परन्तु BA नहीं

(C) BA विद्यमान है परन्तु AB नहीं है (D) AB तथा BA दोनों विद्यमान नहीं हैं

Ans. A

Sol. Since, order of A is 2×3 and order of B is 3×2

Thus, AB and BA exists.

9. $\begin{bmatrix} 1 & -\tan \alpha/2 \\ \tan \alpha/2 & 1 \end{bmatrix}\begin{bmatrix} \tan\alpha/2 & 1 \\ 1 & 0 \end{bmatrix}$ equals :

(A) zero matrix (B) $\sec^2 \alpha.I_2$ (C) I_2 (D) None of these

$\begin{bmatrix} 1 & -\tan \alpha/2 \\ \tan \alpha/2 & 1 \end{bmatrix}\begin{bmatrix} \tan\alpha/2 & 1 \\ 1 & 0 \end{bmatrix}$ बराबर है :

(A) शून्य मैट्रिक्स (B) $\sec^2 \alpha.I_2$ (C) I_2 (D) इनमें से कोई नहीं

Ans. D

Sol. $\begin{bmatrix} 1 & -\tan\dfrac{\alpha}{2} \\ \tan\dfrac{\alpha}{2} & 1 \end{bmatrix}\begin{bmatrix} \tan\dfrac{\alpha}{2} & 1 \\ 1 & 0 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 0 & 1 \\ \sec^2\dfrac{\alpha}{2} & 0 \end{bmatrix}$

10. If $A=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B=\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, then :

(A) AB = BA (B) $AB = B^2$ (C) AB = – BA (D) None of these

यदि $A=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ तथा $B=\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, तो :

(A) AB = BA (B) $AB = B^2$ (C) AB = – BA (D) इनमें से कोई नहीं

Ans. C

Sol. $AB=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

$AB=\begin{bmatrix} i & 0 \\ 0 & -1 \end{bmatrix}$

$$BA = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$BA = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$$

$AB = -BA$

11. If $A = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}$, then AB equals :

यदि $A = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$ तथा $B = \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}$ तो AB बराबर है :

(A) $\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$ (B) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (C) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (D) $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$

Ans. C

Sol. $$AB = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

12. If $A = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, then which of the following statement is true?

यदि $A = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ तो कौनसा कथन सत्य है?

(A) $AB = BA$ (B) $AB = -BA$ (C) $A^2 = BC$ (D) $A^2 = B + C$

Ans. B

Sol. $$AB = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$$

$$BA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$= \begin{bmatrix} i & 0 \\ -i & 0 \end{bmatrix}$$

$AB = -BA$

13. If $A = \begin{bmatrix} -2 & 1 \\ 0 & 3 \end{bmatrix}$ and $f(x) = 2x^2 - 3x$, then $f(A)$ equals :

यदि $A = \begin{bmatrix} -2 & 1 \\ 0 & 3 \end{bmatrix}$ तथा $f(x) = 2x^2 - 3x$, तो $f(A)$ बराबर है :

(A) $\begin{bmatrix} 14 & 1 \\ 0 & -9 \end{bmatrix}$ (B) $\begin{bmatrix} -14 & 1 \\ 0 & 9 \end{bmatrix}$ (C) $\begin{bmatrix} 14 & -1 \\ 0 & 9 \end{bmatrix}$ (D) $\begin{bmatrix} -14 & -1 \\ 0 & -9 \end{bmatrix}$

Ans. C

Sol. $A = \begin{bmatrix} -2 & 1 \\ 0 & 3 \end{bmatrix}$

$$\Rightarrow A^2 = \begin{bmatrix} -2 & 1 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} -2 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 0 & 9 \end{bmatrix}$$

$$\therefore f(A) = 2\begin{bmatrix} 4 & 1 \\ 0 & 9 \end{bmatrix} - 3\begin{bmatrix} -2 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 2 \\ 0 & 18 \end{bmatrix} - \begin{bmatrix} -6 & 3 \\ 0 & 9 \end{bmatrix} = \begin{bmatrix} 14 & -1 \\ 0 & 9 \end{bmatrix}$$

14. If $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$ then $A^2 - 5A + 6I =$

यदि $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$ तो $A^2 - 5A + 6I =$

(A) $\begin{bmatrix} 1 & -1 & -5 \\ -1 & -1 & 4 \\ -3 & -10 & 4 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{bmatrix}$ (C) O (D) I

Ans. B

Sol. $A^2 = A.A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}\begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix}$$

$$\therefore A^2 - 5A + 6I = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} - 5\begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} + 6\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} - \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 5-10+6 & -1-0+0 & 2-5+0 \\ 9-10+0 & -2-5+6 & 5-15+0 \\ 0-5+0 & -1+5+0 & -2+0+6 \end{bmatrix}$$

$$= \left[(1,-1,-3),(-1,-1,-10),(-5,4,4)\right]$$

15. If order of A + B is n × n, then the order of AB is :

(A) n × n (B) n × m (C) m × n (D) not defined

यदि A + B की कोटि n x n हो, तो AB की कोटि होगी :

(A) n × n (B) n × m (C) m × n (D) अपरिभाषित

Ans. A

Sol. If order of A + B is n × n, then the order of both A and B is n × n

Therefore, order of AB is n × n

16. If $A = [a\ b]$, $B = [-b\ -a]$ and $C = \begin{bmatrix} a \\ -a \end{bmatrix}$, then true statement is :

यदि $A = [a\ b]$, $B = [-b\ -a]$ तथा $C = \begin{bmatrix} a \\ -a \end{bmatrix}$, तो सत्य कथन है :

(A) $A = -B$ (B) $A + B = A - B$ (C) $AC = BC$ (D) $CA = CB$

Ans. C

Sol. $AC = \begin{bmatrix} a & b \end{bmatrix}\begin{bmatrix} a \\ -a \end{bmatrix} = \begin{bmatrix} a^2 - ab \end{bmatrix}$

$BC = \begin{bmatrix} -b & -a \end{bmatrix}\begin{bmatrix} a \\ -a \end{bmatrix} = \begin{bmatrix} a^2 - ab \end{bmatrix}$

17. If $A = \begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 2 & -2 \end{bmatrix}$, then $(AB)^T$ is :

यदि $A = \begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix}$ तथा $B = \begin{bmatrix} 3 & 4 \\ 2 & -2 \end{bmatrix}$, तो $(AB)^T$ है :

(A) $\begin{bmatrix} 11 & -2 \\ 5 & -6 \end{bmatrix}$ (B) $\begin{bmatrix} 11 & 5 \\ -2 & -6 \end{bmatrix}$ (C) $\begin{bmatrix} 7 & 1 \\ 0 & -8 \end{bmatrix}$ (D) $\begin{bmatrix} 7 & 0 \\ 1 & -8 \end{bmatrix}$

Ans. C

Sol. $AB = \begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 3 & 4 \\ 2 & -2 \end{bmatrix}$

$AB = \begin{bmatrix} 7 & 0 \\ 1 & -8 \end{bmatrix}$

$(AB)^T = \begin{bmatrix} 7 & 1 \\ 0 & -8 \end{bmatrix}$

18. If A and B are matrices of order $m \times n$ and $n \times m$ respectively, then the order of matrix $B^T (A^T)^T$ is :

(A) $m \times n$ (B) $m \times m$ (C) $n \times n$ (D) Not defined

यदि A तथा B क्रमशः $m \times n$ तथा $n \times m$ कोटि के मैट्रिक्स हों, तो मैट्रिक्स $B^T (A^T)^T$ होगा :

(A) $m \times n$ (B) $m \times m$ (C) $n \times n$ (D) परिभाषित नहीं

Ans. D

Sol. To find the order of $B^T\left(A^T\right)^T$.

Given, A and B are matrices of order $m \times n$ and $n \times m$ respectively.

therefore, order of $A^T = n \times m$.

order of $\left(A^T\right)^T = m \times n$.

order of $B^T = m \times n$.

By the rule of matrix multiplication, number of columns in first matrix must be equal to number of rows in second matrix but in the above case, n is not equal to m.

19. If A, B, C, are three matrices, then $A^T + B^T + C^T$ is :

(A) zero matrix (B) $A + B + C$ (C) $-(A + B + C)$ (D) $(A + B + C)^T$

यदि A, B, C तीन मैट्रिक्स हैं तो $A^T + B^T + C^T$ है :

(A) शून्य मैट्रिक्स (B) $A + B + C$ (C) $-(A + B + C)$ (D) $(A + B + C)^T$

Ans. D

Sol. Here A' is the transpose Matrix also written as A^t, A^{tr}, t_A, A^T

We know transpose respect addition

$$A^T + B^T + C^T = \left(A + B + C\right)^T$$

Hence $\left(A + B + C\right)^T$ is correct.

1. If $A=\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}$ and $B=\begin{bmatrix}-1 & 2\\ -1 & 1\end{bmatrix}$, then correct statement is :

(A) $AB=BA$ (B) $AA^T=A^2$ (C) $AB=B^2$ (D) None of these

यदि $A=\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}$ तथा $B=\begin{bmatrix}-1 & 2\\ -1 & 1\end{bmatrix}$ तो सत्य कथन है :

(A) $AB=BA$ (B) $AA^T=A^2$ (C) $AB=B^2$ (D) इनमें से कोई नहीं

Ans. D

Sol. For $A=\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}$ and $B=\begin{bmatrix}-1 & 2\\ -1 & 1\end{bmatrix}$

$$AB=\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}\begin{bmatrix}-1 & 2\\ -1 & 1\end{bmatrix}=\begin{bmatrix}-5 & 8\\ -5 & 6\end{bmatrix}$$

2. If $A=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$, then AA^T equals :

यदि $A=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$ तो AA^T बराबर है :

(A) $\begin{bmatrix}\cos 2\theta & -\sin 2\theta\\ \sin 2\theta & \cos 2\theta\end{bmatrix}$ (B) $\begin{bmatrix}\cos^2\theta & \sin^2\theta\\ \sin^2\theta & \cos^2\theta\end{bmatrix}$ (C) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ (D) $\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$

Ans. C

Sol. $A^T=\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$

$$AA^T=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

3. If $A=\begin{bmatrix} a & b \\ b & a \end{bmatrix}$, then $|A+A^T|$ equals :

यदि $A=\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ तो $|A+A^T|$ बराबर है :

(A) $4(a^2-b^2)$ (B) $2(a^2-b^2)$ (C) a^2-b^2 (D) $4ab$

Ans. A

Sol. $A=\begin{bmatrix} a & b \\ b & a \end{bmatrix}$

$A^T=\begin{bmatrix} a & b \\ b & a \end{bmatrix}$

$A+A^T=\begin{bmatrix} 2a & 2b \\ 2b & 2a \end{bmatrix}$

$\left|A+A^T\right|=4a^2-4b^2$

$=4\left(a^2-b^2\right)$

4. If $A=\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$, $B=\begin{bmatrix} 3 & 4 \\ 2 & 1 \\ 1 & 3 \end{bmatrix}$, then (AB)' equals :

यदि $A=\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$, $B=\begin{bmatrix} 3 & 4 \\ 2 & 1 \\ 1 & 3 \end{bmatrix}$ तो (AB)' बराबर है :

(A) $\begin{bmatrix} 16 & 22 \\ 23 & 31 \end{bmatrix}$ (B) $\begin{bmatrix} 16 & 23 \\ 22 & 31 \end{bmatrix}$ (C) $\begin{bmatrix} 22 & 31 \\ 16 & 30 \end{bmatrix}$ (D) $\begin{bmatrix} 23 & 16 \\ 31 & 20 \end{bmatrix}$

Ans. A

Sol. $AB=\begin{bmatrix}2 & 3 & 4\\ 3 & 4 & 5\end{bmatrix}\begin{bmatrix}3 & 4\\ 2 & 1\\ 1 & 3\end{bmatrix}$

$AB=\begin{bmatrix}16 & 23\\ 22 & 31\end{bmatrix}$

$(AB)^T=\begin{bmatrix}16 & 22\\ 23 & 31\end{bmatrix}$

5. If $A=\begin{bmatrix}3 & x\\ y & 0\end{bmatrix}$ and $A=A^T$, then :

यदि $A=\begin{bmatrix}3 & x\\ y & 0\end{bmatrix}$ तथा $A=A^T$, तो :

(A) $x=0, y=3$ (B) $x+y=3$ (C) $x=y$ (D) $x=-y$

Ans. C

Sol. $A=A^T$

$\begin{bmatrix}3 & x\\ y & 0\end{bmatrix}=\begin{bmatrix}3 & y\\ x & 0\end{bmatrix}$

$x=y$

6. If $A=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$, then AA' equals :

यदि $A=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$ तो AA' बराबर है :

(A) I (B) A (C) A' (D) 0

Ans. A

Sol. $A=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$

$$A^T = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$$AA^T = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

7. If A is symmetric as well as skew symmetric matrix, then :

(A) A is a diagonal matrix (B) A is a null matrix

(C) A is a unit matrix (D) A is a triangular matrix

यदि A सममित एवं विषम सममित मैट्रिक्स हों, तो-

(A) A विकर्ण मैट्रिक्स होगा (B) A शून्य मैट्रिक्स होगा

(C) A इकाई मैट्रिक्स होगा (D) A एक त्रिभुजाकार मैट्रिक्स होगा

Ans. B

Sol. Since, A is skew-symmetric.

$A^T = -A$

Since, A is symmetric.

$A^T = A$

$\Rightarrow -A = A$

$\Rightarrow 2A = O$

$\Rightarrow A = O$

Hence, A is a null matrix.

8. Let A be a square matrix. Then which of the following is not a symmetric matrix :

माना A वर्ग मैट्रिक्स है तो निम्न में से कौन सा मैट्रिक्स सममित नहीं है :

(A) $A + A'$ (B) AA' (C) $A'A$ (D) $A - A'$

Ans. D

Sol. If a is a square matrix, then and if A' represents its transpose, then $A + A'$ is symmetric and $A - A'$ is skew symmetric.

Hence matrix A can be written as $A = \left(\frac{A + A'}{2}\right) + \left(\frac{A - A'}{2}\right)$

9. For any 2 x 2 matrix A , $A(\text{adj } A) = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$, then |A| equals :

कोटि 2 × 2 के मैट्रिक्स A के लिए, $A(\text{adj } A) = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ तो |A| बराबर है :

(A) 0 (B) 3 (C) 6 (D) 9

Ans. B

Sol. We know that day property of adjoint and inverse of matrix

$A(\text{adj } A) = |A| I$

$$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

By comparing

$|A| = 3$

10. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 0 & 2 \end{bmatrix}$, then the value of adj (adj A) is :

यदि $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 0 & 2 \end{bmatrix}$ तो adj (adj A) का मान होगा :

(A) $|A|^2$ (B) $-2A$ (C) $2A$ (D) A^2

Ans. B

Sol. $\text{adj}(\text{adj } A) = |A|^{n-2} A$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 0 & 2 \end{bmatrix}$$

$|A| = -2, \quad n = 3$

$\text{adj}(\text{adj } A) = |A| A$

$= -2A$

1. If $A=\begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$ and $A(\text{adj } A)=k\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then k equals :

यदि $A=\begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$ तथा $A(\text{adj } A)=k\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, तो k बराबर है :

(A) $\sin x \cos x$ (B) 1 (C) 2 (D) 3

Ans. B

Sol. $A=\begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$

$|A| = 1$

$A(\text{adj } A) = |A| I$

$k = 1$

2. If $A=\begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 1 \\ 2 & 1 & 2 \end{bmatrix}$, then A (adj A) equals :

(A) $\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$ (B) $-\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & 0 & 9 \\ 0 & 9 & 0 \\ 9 & 0 & 0 \end{bmatrix}$ (D) None of these

यदि $A=\begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 1 \\ 2 & 1 & 2 \end{bmatrix}$ तो A (adj A) बराबर है :

(A) $\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$ (B) $-\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & 0 & 9 \\ 0 & 9 & 0 \\ 9 & 0 & 0 \end{bmatrix}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $A=\begin{bmatrix}1 & 2 & 3\\ 0 & 3 & 1\\ 2 & 1 & 2\end{bmatrix}$

$|A| = 5+2(2-9) = -9$

$A(\text{adj } A) = |A|\, I$

$$= -9\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$= -\begin{bmatrix}9 & 0 & 0\\ 0 & 9 & 0\\ 0 & 0 & 9\end{bmatrix}$$

3. $(\text{adj } A^T) - (\text{adj } A)^T$ equals :

(A) $2|A|$ (B) $2|A|I$ (C) zero matrix (D) Unit matrix

$(\text{adj } A^T) - (\text{adj } A)^T$ बराबर है :

(A) $2|A|$ (B) $2|A|I$ (C) शून्य मैट्रिक्स (D) इकाई मैट्रिक्स

Ans. C

Sol. $\left(\text{adj } A^T\right) = \left(\text{adj } A\right)^T$

$\left(\text{adj } A^T\right) - \left(\text{adj } A\right)^T = 0$

4. If $A=\begin{bmatrix}1 & 3 & 5\\ 3 & 5 & 1\\ 5 & 1 & 3\end{bmatrix}$ then adj A equals :

(A) $\begin{bmatrix}14 & -4 & -22\\ -4 & -22 & 14\\ -22 & 14 & -4\end{bmatrix}$ (B) $\begin{bmatrix}-14 & 4 & 22\\ 4 & 22 & -14\\ 22 & -14 & 4\end{bmatrix}$

(C) $\begin{bmatrix} 14 & 4 & 22 \\ 4 & 22 & 14 \\ 22 & 14 & 4 \end{bmatrix}$ (D) None of these

यदि $A = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 5 & 1 \\ 5 & 1 & 3 \end{bmatrix}$ तो adj A बराबर है :

(A) $\begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$ (B) $\begin{bmatrix} -14 & 4 & 22 \\ 4 & 22 & -14 \\ 22 & -14 & 4 \end{bmatrix}$

(C) $\begin{bmatrix} 14 & 4 & 22 \\ 4 & 22 & 14 \\ 22 & 14 & 4 \end{bmatrix}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\begin{array}{|c|c|c|} C_{11} = 14 & C_{21} = -4 & C_{31} = -22 \\ C_{12} = -4 & C_{22} = -22 & C_{32} = 14 \\ C_{13} = -22 & C_{23} = 14 & C_{33} = -4 \end{array}$

$$\text{adj } A = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}^T$$

$$= \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$$

5. If A is a square matrix of order n, then the value of |adj A| is :

यदि A वर्ग मैट्रिक्स है जिसकी कोटि n है, तो |adj A| का मान होगा :

(A) $|A|^{n-1}$ (B) $|A|^n$ (C) $|A|^{n+1}$ (D) $|A|^{n+2}$

Ans. A

Sol. |Adj A| of a square matrix of nth order, is given by $|A|^{n-1}$ where |A| is determinant of square matrix of A of nth order.

6. The adjoint of symmetric matrix is :

(A) symmetric matrix (B) skew-symmetric matrix

(C) diagonal matrix (D) None of these

सममित आव्यूह के सहखण्डज होता है :

(A) सममित मैट्रिक्स (B) विषम सममित मैट्रिक्स

(C) विकर्ण मैट्रिक्स (D) इनमें से कोई नहीं

Ans. A

Sol. Let A is symmetric matrix then :

$\left(A^T\right) = A$

We know the property :

$\text{adj}\left(A^T\right) = \left(\text{adj}A\right)^T$

$\text{adj}\left(A\right) = \left(\text{adj } A\right)^T \left\{A^T = A\right\}$

$\left(\text{adj}\left(A\right)\right)^T = \text{adj}\left(A\right) \rightarrow \text{symmetric}$

7. The inverse matrix of $\begin{bmatrix} 4 & 7 \\ 1 & 2 \end{bmatrix}$ is :

मैट्रिक्स $\begin{bmatrix} 4 & 7 \\ 1 & 2 \end{bmatrix}$ का प्रतिलोम मैट्रिक्स है :

(A) $\begin{bmatrix} 2 & -7 \\ -1 & 4 \end{bmatrix}$ (B) $\begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$ (C) $\begin{bmatrix} -2 & 7 \\ 1 & -4 \end{bmatrix}$ (D) $\begin{bmatrix} -2 & 1 \\ 7 & -4 \end{bmatrix}$

Ans. A

Sol. Let $A = \begin{bmatrix} 4 & 7 \\ 1 & 2 \end{bmatrix}$

$\text{adj } A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$

$|A| = 1$

$$\therefore A^{-1} = \frac{\text{adj } A}{|A|} = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$$

8. If $A = \begin{bmatrix} 2 & 3 \\ 1 & 3 \end{bmatrix}, B = \begin{bmatrix} 4 & 6 \\ 2 & 3 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then invertible matrices are :

(A) A and B (B) B and C (C) A and C (D) All

यदि $A = \begin{bmatrix} 2 & 3 \\ 1 & 3 \end{bmatrix}, B = \begin{bmatrix} 4 & 6 \\ 2 & 3 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ तो व्युत्क्रमणीय मैट्रिक्स होंगे :

(A) A तथा B (B) B तथा C (C) A तथा C (D) सभी

Ans. C

Sol. $|A| = 3, |B| = 0, |C| = 1$

B is not invertible matrix

9. Matrix $\begin{bmatrix} \lambda & -1 & 4 \\ -3 & 0 & 1 \\ -1 & 1 & 2 \end{bmatrix}$ is not invertible if :

मैट्रिक्स $\begin{bmatrix} \lambda & -1 & 4 \\ -3 & 0 & 1 \\ -1 & 1 & 2 \end{bmatrix}$ व्युत्क्रमणीय नहीं है यदि :

(A) $\lambda = -15$ (B) $\lambda = -17$ (C) $\lambda = -16$ (D) $\lambda = -18$

Ans. B

Sol. $\begin{vmatrix} \lambda & -1 & 4 \\ -3 & 0 & 1 \\ -1 & 1 & 2 \end{vmatrix} = 0$

$$-\lambda + (-6+1) + 4(-3) = 0$$

$$\lambda = -17$$

10. If $A=\begin{bmatrix}2&0&0\\0&2&0\\0&0&2\end{bmatrix}$, then value of A^{-1} is :

(A) $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$ (B) $\begin{bmatrix}\frac{1}{2}&0&0\\0&\frac{1}{2}&0\\0&0&\frac{1}{2}\end{bmatrix}$ (C) $\begin{bmatrix}-2&0&0\\0&-2&0\\0&0&-2\end{bmatrix}$ (D) None of these

यदि $A=\begin{bmatrix}2&0&0\\0&2&0\\0&0&2\end{bmatrix}$ तो A^{-1} का मान होगा :

(A) $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$ (B) $\begin{bmatrix}\frac{1}{2}&0&0\\0&\frac{1}{2}&0\\0&0&\frac{1}{2}\end{bmatrix}$ (C) $\begin{bmatrix}-2&0&0\\0&-2&0\\0&0&-2\end{bmatrix}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $A=\begin{bmatrix}2&0&0\\0&2&0\\0&0&2\end{bmatrix}$

$|A| = 8$

$\text{adj } A=\begin{bmatrix}4&0&0\\0&4&0\\0&0&4\end{bmatrix}$

$$A^{-1} = \frac{\text{adj } A}{|A|} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$$

11. For any square matrix A, which statement is wrong :

(A) $(\text{adj } A)^{-1} = \text{adj } (A^{-1})$ (B) $(A^T)^{-1} = (A^{-1})^T$

(C) $(A^3)^{-1} = (A^{-1})^3$ (D) None of these

किसी वर्ग मैट्रिक्स A के लिए कौन सा कथन असत्य है :

(A) $(\text{adj } A)^{-1} = \text{adj } (A^{-1})$ (B) $(A^T)^{-1} = (A^{-1})^T$

(C) $(A^3)^{-1} = (A^{-1})^3$ (D) इनमें से कोई नहीं

Ans. D

Sol. $\text{adj}(AB) = \text{Adj}(B)\,\text{Adj}(A)$

$\text{adj}(AA^{-1}) = \text{adj}(A^{-1})\,\text{Adj}(A)$

$\text{adj}(I) = \text{Adj}(A^{-1})\,\text{Adj}(A)$

$$\frac{I}{(\text{Adj } A)} = \text{Adj}\left(A^{-1}\right)$$

$$\left(\text{Adj } A\right)^{-1} = \text{Adj}\left(A^{-1}\right)$$

This option is correct.

Option : B

$$\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$$

$$A^T\left(A^{-1}\right)^T = \left(A^{-1}A\right)^T = I^T = I \rightarrow (1)$$

Parallelly

$$\left(A^{-1}\right)^T\left(A^T\right) = \left(AA^{-1}\right)^T = I^T = I \rightarrow (2)$$

From (1) and (2)

$$\left(A^{-1}\right)^T (A)^T = I$$

$\Rightarrow \left(A^T\right)^{-1} = \left(A^{-1}\right)^T$

$\left(A^3\right)^{-1} = \left(A^{-1}\right)^3$

$AA^{-1} = I$

$\Rightarrow (AA^{-1})^3 = I^3$

$\Rightarrow A^3(A^{-1})^3 = I$

$\Rightarrow (A^T)^{-1} = (A^{-1})^T$

Option-A, B, C are true.

12. If $A = \begin{bmatrix} 1 & 2 \\ -4 & -1 \end{bmatrix}$, then $A^{-1} =$

यदि $A = \begin{bmatrix} 1 & 2 \\ -4 & -1 \end{bmatrix}$, तो $A^{-1} =$

(A) $\frac{1}{7}\begin{bmatrix} 1 & 2 \\ -4 & -1 \end{bmatrix}$ (B) $\begin{bmatrix} -1 & -2 \\ 4 & 1 \end{bmatrix}$ (C) $\frac{1}{9}\begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$ (D) $\frac{1}{7}\begin{bmatrix} -1 & -2 \\ 4 & 1 \end{bmatrix}$

Ans. D

Sol. $A = \begin{bmatrix} 1 & 2 \\ -4 & -1 \end{bmatrix}$

$|A| = 7$

$\text{adj } A = \begin{bmatrix} -1 & -2 \\ 4 & 1 \end{bmatrix}$

$A^{-1} = \frac{\text{adj } A}{|A|} = \frac{1}{7}\begin{bmatrix} -1 & -2 \\ 4 & 1 \end{bmatrix}$

13. Which of the following matrix is inverse of itself :

निम्न में से कौनसी मैट्रिक्स स्वयं की प्रतिलोम है :

(A) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

Ans. B

Sol. Inverse of unit matrix also unit matrix.

1. If $A=\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$ and $n \in N$, then A^n is equal to :

(A) 2^nA (B) $2^{n-1}A$ (C) nA (D) None of these

यदि $A=\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$ तथा $n \in N$, तो A^n बराबर है :

(A) 2^nA (B) $2^{n-1}A$ (C) nA (D) इनमें से कोई नहीं

Ans. B

Sol. $A=\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$

$$A^2=\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}=\begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}=2A$$

$$A^3=A^2.A=2A.A=2\left(A^2\right)=2.2A=2^{(3-1)}A$$

Then in general $A^n=2^{n-1}A$

2. If $E(\theta)=\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$ then $E(\alpha)\,E(\beta)$ is equal to :

यदि $E(\theta)=\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$ तो $E(\alpha)\,E(\beta)$ बराबर है :

(A) E(O) (B) $E(\alpha\beta)$ (C) $E(\alpha+\beta)$ (D) $E(\alpha-\beta)$

Ans. C

Sol. $$E(\alpha)E(\beta)=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}\begin{bmatrix}\cos\beta & \sin\beta\\ -\sin\beta & \cos\beta\end{bmatrix}$$

$$=\begin{bmatrix}\cos(\alpha+\beta) & \sin(\alpha+\beta)\\ -\sin(\alpha+\beta) & \cos(\alpha+\beta)\end{bmatrix}$$

3. If $A = [a_{ij}]$ is scalar matrix of order n x n such that $a_{ij} = k$ for all i, then $|A|$ equals :

यदि $A = [a_{ij}]$ अदिश मैट्रिक्स की कोटि $n \times n$ हो, एवं $a_{ij} = k$ सभी i के लिऐ, तो $|A|$ बराबर है :

(A) nk (B) $n + k$ (C) n^k (D) k^n

Ans. D

Sol. For scalar matrix $\begin{bmatrix} K & 0 & 0 & 0 & - & 0 \\ 0 & K & 0 & 0 & - & 0 \\ 0 & 0 & K & - & - & 0 \\ - & - & - & K & - & 0 \\ - & - & - & - & K & 0 \\ 0 & 0 & 0 & 0 & 0 & K \end{bmatrix}_{n\times n}$

$= K^n$

4. If $D = \text{diag}(d_1, d_2, ..., d_n)$, then D^n equals :

(A) D (B) $\text{diag}(d_1^{n-1}, d_2^{n-1}, ...d_n^{n-1})$

(C) $\text{diag}(d_1^n, d_2^n, ..., d_n^n)$ (D) None of these

यदि $D = \text{diag}(d_1, d_2, ..., d_n)$, तो D^n बराबर है :

(A) D (B) $\text{diag}(d_1^{n-1}, d_2^{n-1}, ...d_n^{n-1})$

(C) $\text{diag}(d_1^n, d_2^n, ..., d_n^n)$ (D) इनमें से कोई नहीं

Ans. C

Sol. $A = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$

then $A^2 = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}\begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$

$= \begin{bmatrix} d_1^2 & 0 & 0 \\ 0 & d_2^2 & 0 \\ 0 & 0 & d_3^2 \end{bmatrix}$

$$A^3 = A^2 \cdot A = \begin{bmatrix} d_1^3 & 0 & 0 \\ 0 & d_2^3 & 0 \\ 0 & 0 & d_3^3 \end{bmatrix}$$

Hence continuing like this

5. If $A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then :

(A) adj A = A (B) adj A = A^{-1} (C) $A^{-1} = -A$ (D) None of these

यदि $A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ तो :

(A) adj A = A (B) adj A = A^{-1} (C) $A^{-1} = -A$ (D) इनमें से कोई नहीं

Ans. B

Sol. $A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$|A| = \cos^2\theta + \sin^2\theta = 1$

$A^{-1} = \dfrac{\text{adj } A}{|A|}$

$A^{-1} = \text{adj } A$

6. If A is orthogonal matrix, then |A| equals :

यदि A लाम्बिक मैट्रिक्स हो, तो |A| बराबर है :

(A) 0 (B) 1 (C) – 1 (D) ±1

Ans. D

Sol. Given that matrix A is orthogonal matrix

Orthogonal matrix is a square

matrix with real entities whose

columns and rows are orthogonal unit vectors

$\Rightarrow AA^T = I$

Apply det on both sides, we get $|A| \times |A^T| = |I|$

$\Rightarrow |A|^2 = 1$

$\Rightarrow |A| = \pm 1$

Therefore the correct option is D

7. If A is any skew- symmetric matrix of odd orders, then |A| equals :

(A) – 1 (B) 0 (C) 1 (D) None of these

यदि A कोई विषम सममित मैट्रिक्स है जो कि विषम कोटि का है, तो |A| बराबर है :

(A) – 1 (B) 0 (C) 1 (D) इनमें से कोई नहीं

Ans. B

Sol. Since, A is skew-symmetric.

$A = -A$

We know that $|kA| = k^n|A|$ where n is order of matrix

$\therefore \ |A'| = (-1)^n |A|$

$\Rightarrow |A| = (-1)^n |A|$

$\Rightarrow \left(1-(-1)^n\right)|A| = 0$

If n is odd, then $(1 - (-1)^n) \neq 0$

$\Rightarrow |A| = 0$

8. If A and B are square matrices such that AB = B and BA = A, then $A^2 + B^2$ is equal to :

यदि A तथा B वर्ग मैट्रिक्स इस प्रकार से है कि AB = B तथा BA = A, तो $A^2 + B^2$ बराबर है :

(A) 2AB (B) 2BA (C) A + B (D) AB

Ans. C

Sol. $A^2 + B^2 = AA + BB$

$= A(BA) + B(AB)$ (Given AB = B and BA = A)

$=(AB)A+(BA)B$

$=BA+AB$

$=A+B$

9. If $D = diag(d_1, d_2, d_3, ...,d_n)$, where $d_i \neq 0$ for all $i = 1,2,....,n$; then D^{-1} is equal to :

(A) D (B) I_n

(C) diag $(d_1^{-1}, d_2^{-1}, ...,d_n^{-1})$ (D) None of these

यदि $D = diag(d_1, d_2, d_3, ...,d_n)$, जहाँ $d_i \neq 0$ सभी $i = 1,2,....,n$ के लिए ; तो D^{-1} बराबर है :

(A) D (B) I_n

(C) diag $(d_1^{-1}, d_2^{-1}, ...,d_n^{-1})$ (D) इनमें से कोई नहीं

Ans. C

Sol. Given $D= diag [d_1, d_2, d_3]$

It is also given that $d_1 \neq 0, d_2 \neq 0, d_3 \neq 0$

$$\begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$$

A diagonal matrix $D = diag(d_1, d_2, ...d_n)$ is invertible iff all diagonal entries are non-zero,

i.e. $di \neq 0$ for $1 \leq 0 \leq n$

If D is invertible then $D^{-1} = diag(d1^{-1}, ...dn^{-1})$

By the Inverting Diagonal matrices theorem, which states that

Here, it is given that $d_1 \neq 0, d_2 \neq 0, d_3 \neq 0$

$\therefore$ D is invertible

$\Rightarrow D^{-1} = diag\left[d_1^{-1}, d_2^{-1}, d_3^{-1}\right]$

Hence Proved.

10. If A is invertible matrix, then det (A^{-1}) equals :

(A) det A (B) 1/det A (C) 1 (D) None of these

यदि A व्युत्क्रमणीय मैट्रिक्स हो, तो det (A^{-1}) बराबर है :

(A) det A (B) 1/det A (C) 1 (D) इनमें से कोई नहीं

Ans. B

Sol. We know

$AA^{-1} = I$

$\Rightarrow \left|AA^{-1}\right| = |I|$

$\Rightarrow |A|\ \left|A^{-1}\right| = 1$

$\Rightarrow \left|A^{-1}\right| = \frac{1}{|A|}$

$\Rightarrow \left|A^{-1}\right| = |A|^{-1}$

11. If A and B are non- zero square matrices of the same order such that AB = O, then :

(A) adj A = O or adj B = O (B) adj A = O and adj B = O

(C) |A| = 0 or |B| = 0 (D) None of these

यदि A तथा B अशून्य वर्ग मैट्रिक्स समान कोटि वाले हों, एवं AB = O, तो :

(A) adj A = O या adj B = O (B) adj A = O तथा adj B = O

(C) |A| = 0 या |B| = 0 (D) इनमें से कोई नहीं

Ans. C

Sol. $|AB| = |O|$

$|A||B| = O$

$|A| = 0$ or $|B| = 0$

12. If $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $(aI_2 + bA)^2 = A$, then :

यदि $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ तथा $(aI_2 + bA)^2 = A$, तो :

(A) $a = b = \sqrt{2}$ (B) $a = b = 1/\sqrt{2}$ (C) $a = b = \sqrt{3}$ (D) $a = b = 1/\sqrt{3}$

Ans. B

Sol. In the question we were given to find the value of a and b by considering some conditions

$$A^2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$$

$$(al + bA)^2 = (al + bA)(al + bA)$$

$$a^2l^2 + ablA + baAl + b^2A^2$$

$$a^2l + 2abA + b^2A^2$$

$$a^2l + 2abA - b^2l$$

$$(a^2 - b^2)1 + 2abA$$

$$\begin{bmatrix} a^2 - b^2 & 0 \\ 0 & a^2 - b^2 \end{bmatrix} + 2ab\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$=> \begin{bmatrix} a^2 - b^2 & 2ab \\ -2ab & a^2 - b^2 \end{bmatrix}$$

By solving we get $a = b = \pm\frac{1}{\sqrt{2}}$

13. If $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ and $B = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$, then AB is equal to :

यदि $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ तथा $B = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$ तो AB बराबर है :

(A) A (B) B (C) I (D) 0

Ans. D

Sol. $AB=\begin{bmatrix}0 & c & -b\\ -c & 0 & a\\ b & -a & 0\end{bmatrix}\begin{bmatrix}a^2 & ab & ac\\ ab & b^2 & bc\\ ac & bc & c^2\end{bmatrix}$

$AB=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}=0$

14. If $M=\begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}$ and $M^2-\lambda M-I_2=0$, then λ equals :

यदि $M=\begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}$ तथा $M^2-\lambda M-I_2=0$, तो λ बराबर है :

(A) -2 (B) 2 (C) -4 (D) 4

Ans. D

Sol. $M=\begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}$ and $M^2-\lambda M-I_2=O$

$M^2-\lambda M-I_2=O$

$\Rightarrow \lambda M=M^2-I_2$

$\Rightarrow \lambda M=\begin{bmatrix}5 & 8\\ 8 & 13\end{bmatrix}-\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

$\Rightarrow \lambda M=\begin{bmatrix}4 & 8\\ 8 & 12\end{bmatrix}=4M$

$\therefore \lambda=4$

15. If $A=\begin{bmatrix}1 & 3\\ 3 & 4\end{bmatrix}$ and $A^2-kA-5I_2=O$, then the value of k is :

यदि $A=\begin{bmatrix}1 & 3\\ 3 & 4\end{bmatrix}$ तथा $A^2-kA-5I_2=O$, तो k का मान है :

(A) 3 (B) 5 (C) 7 (D) –7

Ans. B

Sol. Given, $A=\begin{bmatrix}1 & 3\\ 3 & 4\end{bmatrix}$

Now, $A^2-5I_2=\begin{bmatrix}1 & 3\\ 3 & 4\end{bmatrix}\begin{bmatrix}1 & 3\\ 3 & 4\end{bmatrix}-5\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

$=\begin{bmatrix}1+9 & 3+12\\ 3+12 & 9+16\end{bmatrix}-\begin{bmatrix}5 & 0\\ 0 & 5\end{bmatrix}$

$=\begin{bmatrix}5 & 15\\ 15 & 20\end{bmatrix}=5\begin{bmatrix}1 & 3\\ 3 & 4\end{bmatrix}$

$\Rightarrow A^2-5I_2=5A$

But it is given that $A^2-5I_2=kA$

$k = 5$.

16. Let $F(\alpha)=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$, where $\alpha \in R$. Then $(F(\alpha))^{-1}$ is equal to :

(A) $F(-\alpha)$ (B) $F(\alpha^{-1})$ (C) $F(2\alpha)$ (D) None of these

माना कि $F(\alpha)=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$, जहाँ $\alpha \in R$ तो $(F(\alpha))^{-1}$ का मान है :

(A) $F(-\alpha)$ (B) $F(\alpha^{-1})$ (C) $F(2\alpha)$ (D) इनमें से कोई नहीं

Ans. A

Sol. Given, $F(\alpha)=A=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$

$\Rightarrow |A|=\cos^2\alpha+\sin^2\alpha=1$

$$\text{adj } A = \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} = f(-\alpha)$$

17. If a matrix A is such that $3A^3 + 2A^2 + 5A + I = 0$, then A^{-1} is equal to :
(A) $-(3A^2 + 2A + 5I)$ (B) $(3A^2 + 2A + 5I)$ (C) $(3A^2 - 2A - 5I)$ (D) None of these

यदि A एक आव्यूह इस प्रकार है कि $3A^3 + 2A^2 + 5A + I = 0$, तो A^{-1} का मान है :
(A) $-(3A^2 + 2A + 5I)$ (B) $(3A^2 + 2A + 5I)$ (C) $(3A^2 - 2A - 5I)$ (D) इनमें से कोई नहीं

Ans. A

Sol. $3A^3 + 2A^2 + 5A + I = 0$

$3A^3 + 2A^2 + 5A + AA^{-1} = 0$

$A^{-1} = 3A^2 - 2A - 5$

18. For the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix}$, which of the following is correct :

आव्यूह $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix}$ के लिए, निम्न में से कौनसा कथन सही है :

(A) $A^3 + 3A^2 - I = O$ (B) $A^3 - 3A^2 - I = O$ (C) $A^3 + 2A^2 - I = O$ (D) $A^3 - A^2 + I = O$

Ans. B

Sol. Now, $A^2 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 6 & 2 \\ 3 & 4 & 1 \end{bmatrix}$

$\Rightarrow A^3 = A^2 \cdot A$

$$= \begin{bmatrix} 2 & 3 & 1 \\ 5 & 6 & 2 \\ 3 & 4 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 7 & 9 & 3 \\ 15 & 19 & 6 \\ 9 & 12 & 4 \end{bmatrix}$$

Now, $A^3 - 3A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\Rightarrow A^3 - 3A^2 = 1$

$\Rightarrow A^3 - 3A^2 - 1 = 0$

19. If matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & 6 & 7 \end{bmatrix}$ and its inverse is denoted by $A^{-1} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, then the value of a_{23} is equal to :

यदि आव्यूह $A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & 6 & 7 \end{bmatrix}$ तथा इसका प्रतिलोम आव्यूह $A^{-1} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ से प्रदर्शित होता है, तो a_{23} बराबर है :

(A) $\frac{21}{20}$ (B) $\frac{1}{5}$ (C) $-\frac{2}{5}$ (D) $\frac{2}{5}$

Ans. D

Sol. Given, $A = \begin{vmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & 6 & 7 \end{vmatrix}$

$|A| = 1(28-30) - 1(18)$

$|A| = -2 - 18$

$|A| = -20$

The cofactor matrix is $C = \begin{vmatrix} -2 & -28 & 18 \\ -6 & 7 & -6 \\ -4 & -8 & 4 \end{vmatrix}$

Thus, the adjoint of A is C^T

Thus $A^{-1} = \frac{1}{-20}\begin{vmatrix} -2 & -6 & 4 \\ -28 & 7 & -8 \\ 18 & -6 & 4 \end{vmatrix}$

Therefore, $a_{23} = \frac{-8}{-20} = \frac{2}{5}$

20. If $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -3 \\ 2 & 1 & 0 \end{bmatrix}$ and B = (adj A), and C = 5A, then $\frac{|adjB|}{|C|}$ is equal to :

यदि $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -3 \\ 2 & 1 & 0 \end{bmatrix}$ तथा B = (adj A), तथा C = 5A, तो $\frac{|adjB|}{|C|}$ बराबर है :

(A) 5 (B) 25 (C) –1 (D) 1

Ans. D

Sol. $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -3 \\ 2 & 1 & 0 \end{bmatrix}$

$B = \text{adj } A$

$|\text{adj } B| = |\text{adj}(\text{adj } A)| = (A)^{(n-1)^2}$

$= |A|^{(3-1)^2} = |A|^4$

$|C| = |5A| = 125|A|$

$|A| = 5$

21. If $P = \begin{pmatrix} i & 0 & -i \\ 0 & -i & i \\ -i & i & 0 \end{pmatrix}$ and $Q = \begin{pmatrix} -i & i \\ 0 & 0 \\ i & -i \end{pmatrix}$, then PQ is equal to :

यदि $P=\begin{pmatrix} i & 0 & -i \\ 0 & -i & i \\ -i & i & 0 \end{pmatrix}$ तो $Q=\begin{pmatrix} -i & i \\ 0 & 0 \\ i & -i \end{pmatrix}$ तो PQ बराबर है :

(A) $\begin{pmatrix} -2 & 2 \\ 1 & -1 \\ 1 & -1 \end{pmatrix}$ (B) $\begin{pmatrix} 2 & -2 \\ -1 & 1 \\ -1 & 1 \end{pmatrix}$ (C) $\begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix}$ (D) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Ans. B

Sol. $PQ=\begin{bmatrix} i & 0 & -i \\ 0 & -i & i \\ -i & i & 0 \end{bmatrix}\begin{bmatrix} -i & i \\ 0 & 0 \\ i & -i \end{bmatrix}$

$=\begin{bmatrix} 2 & -2 \\ -1 & 1 \\ -1 & 1 \end{bmatrix}$

22. If A, B are symmetric matrices of the same order then (AB – BA) is :

(A) symmetric matrix (B) skew symmetric matrix

(C) null matrix (D) unit matrix

यदि A तथा B समान कोटि के दो सममित आव्यूह है, तो (AB – BA) बराबर है :

(A) सममित आव्यूह (B) विषम सममित आव्यूह

(C) रिक्त आव्यूह (D) इकाई आव्यूह

Ans. B

Sol. Let A and B be symmetric matrices.

Then $A'=A$ and $B'=B$

Now,

$(AB-BA)'=(AB)'-(BA)'$

$=(B'A')-(A'B')$

$=(BA-AB)$

$=-(AB-BA)$

Thus,

$(AB - BA)' = -(AB - BA)$.

Hence, $(AB - BA)'$ is skew-symmetric.